FEDERAL SPECIFICATION

INSULATING FLUID, ELECTRICAL, SILICONE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers high fire point electrical insulating fluids of the silicone type.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-290 - Packaging, Packing and Marking of Petroleum and Related Products.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other specifications required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston; New York; Philadelphia; Washington, DC; Atlanta; Chicago; Kansas City, MO; Denver; Forth Worth; Houston; San Francisco; Los Angeles; and Seattle, WA.

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply:

Underwriters' Laboratories, Inc.

UL Document 340

American Society for Testing and Materials (ASTM) Standards:

D92 -Standard Method of Test for Flash and Fire Points by Cleveland Open Cup.
D97 -Standard Methods of Test for Cloud and Pour Points.
D445 -Standard Method of Test for Viscosity, Glass Capillary.
D877 -(Referee Test) Standard Method of Test for Dielectric Breakdown Voltage of Insulating Liquids Using Disk Electrodes.
D923 -Sampling Electrical Insulating Fluids.
D924 -Standard Method of Test for Power Factor and Dielectric Constant of Electrical Insulating Liquids.
D974 -Standard Method of Test for Neutralization Number.
D1169 -Standard Method of Test for Specific Resistance (Resistivity) of Electrical Insulating Liquids.
D1209 -Standard Method of Test for Color, APHA.
D1298 -Standard Method of Test for Specific Gravity by Hydrometer.
D1533 -Standard Method of Test for Water in Insulating Liquids (modified Karl Fischer Method substituting formamide for chloroform).

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103).

3. REQUIREMENTS

3.1 Materials. The insulating fluid shall be a high fire point, electrical insulating and cooling liquid of the silicone type.

3.2 Properties. The electrical insulating fluid shall be clear, free of suspended matter and sediment and shall conform to the requirements specified in Table I when tested as specified in 4.3.2.

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TABLE I. Requirements of Electrical Insulating Fluid

Properties	Limits
Fire point	340°C min
Flash point	300°C min
Pour point	-40°C max
Color	15 max
Viscosity at 25°C, centistoke	47.5 - 52.5
Specific gravity at 25°C	0.957 - 0.963
Neutralization number, BCP	0.01 max
Water content, ppm	50 max
Dielectric strength, 0.1 inch gap at 25°C, 3000 volts/sec. rate of rise, volts/mil	350 min
Dissipation factor (power factor) at 23°C, 60 Hz	1×10^{-4} max
Volume resistivity, ohm-cm	1×10^{14} min
U-L fire hazard classification	4 - 5

3.3 Net content. The average net content per unit container shall be not less than specified.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, in the contractor order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Sampling for examination of filled containers. Take a random sample of filled containers from each lot in accordance with MIL-STD-105, at inspection level II and acceptable quality level (AQL) = 2.5 percent defective.

4.2.2 Sampling for quality conformance tests. Take samples for tests in accordance with ASTM Method D923.

4.3 Examination of filled containers. Examine samples taken in accordance

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with 4.2.1 for compliance with MIL-STD-290 with regard to fill, closure, sealing, leakage, packaging, packing and marking of the containers. Reject any container having one or more defects or under the required fill. If the number of defective or underfilled containers exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, reject the lot represented by the sample."

4.4 Testing of the end item. The applicable methods of testing specified in the ASTM Standards as listed in Table II shall be followed for each lot. The lot size shall be expressed in gallons of electrical insulating fluid. As specified in Table II, tests shall be performed on the composite or on the sample units. The sample unit for testing shall be one gallon of insulating fluid. The sample size shall be as specified below. The composite sample for testing shall be a one gallon quantity obtained by throroughly mixing equal portions from sample selected at random throughout the lot in the sample size frequency indicated below. The procedure for obtaining the sample units and composite shall be as specified in ASTM D923. Care shall be exercised to prevent contamination or alteration of sample unites and composite during sampling, storage, and testing. All test reports shall contain the individual values utilized in expressing the final results. The lot shall not be acceptable if any sample unit or the composite fails to meet any test requirement specified.

Lot size (gallons)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

TABLE II. Instructions for Testing of the End Item

Characteristic	Specification Reference: Requirement	Specification Reference: Test Method	Requirements applicable to: Sample Unit	Requirements applicable to: Comp. Sample	Number of determinations per unit	Results reported as
Fire point	Table I	D92		X	1	Pass or fail
Flash point	Table I	D92		X	1	Pass or fail
Pour point	Table I	D97		X	1	Pass or fail
Color, APHA	Table I	D1209		X	1	Pass or fail
Viscosity at 25°C	Table I	D445		X	1	Pass or fail
Specific gravity	Table I	D1298		X	1	Pass or fail

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TABLE II. Instructions for Testing of the End Items (Con.)

Characteristic	Specification Reference: Requirement	Specification Reference: Test Method	Requirements applicable to: Sample Unit	Requirements applicable to: Comp. Sample	Number of determinations per unit	Results reported as
Neutralization number	Table I	D974		X	1	Pass or fail
Water content	Table I	D1533	X		1	Pass or fail
Dielectric strength	Table I	D877	X		3	Average of 3 determination to nearest 10 volts
Dissipation factor (power factor)	Table I	D924	X		1	Pass or fail
Resistivity	Table I	D1169	X		1	Pass or fail

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. The electrical insulating fluid shall be furnished in 55-gallon quantities. Packaging and packing for each level of protection shall be in accordance with the applicable requirements of MIL-STD-290 (see 6.2).

5.2 Marking. Unit containers and shipping containers shall be marked for shipment in accordance with MIL-STD-290.

6. NOTES

6.1 Intended. The use of electrical insulating fluid is generally regulated by the engineering design of the electrical apparatus. The electrical insulating fluid covered by this specification is a silicone liquid intended for use as an insulating and cooling medium in electrical transformers when a high point fluid is required.

6.1.1 Precaution in use. The electrical insulating fluid covered by this specification is not interchangeable with electrical insulating oils covered in VV-I-530 for any specified electrical equipment or component. The electrical insulating fluid must never be mixed with the insulating oil. When transferring electrical insulating fluid from its original container, great care should be taken to prevent contamination with moisture, dirt, and foreign matter. These impurities cause rapid deterioration of the dielectric strength. Care should also be taken in sealing partially filled containers.

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6.1.2 Storage. Care should be taken to ensure that moisture cannot enter the containers. Containers shall preferably be stored indoors.

6.1.3 Toxicity. Silicone transformer liquid is essentially non-toxic and non-irritating although temporary discomfort may result if rubbed into the eye.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1).

MILITARY CUSTODIANS:

Army - AR
Air Force - 68

Review activities:

Army - MS, MR
Air Force - 68
GS

User activities:

Army - CE, CR

Preparing activity:

Army - AR

CIVIL AGENCY INTEREST:

DOT-ACO
INT-BPA
GSA-FSS

Project Number: 9160-0062

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain copies and other documents referenced herein. Price cents each.

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